

Sumit Banik | Curriculum Vitae

Postdoctoral Researcher, Stanford University & SLAC, California 

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Personal Information

	Age	29
	Nationality	Indian
	Residency	United States of America
	Languages	English (Professional), Bengali (Native), Hindi (Fluent), German (Beginner)

Employment

	Postdoctoral Researcher Stanford University & SLAC National Accelerator Laboratory Full-time (100%), average weekly workload 40 hours, no teaching duties.	<i>Jan 2026 – Now</i> California 
	Postdoctoral Researcher University of Zürich & Paul Scherrer Institut Full-time (100%), average weekly working time 42 hours, including teaching duties every semester depending on availability.	<i>Mar 2023 – Dec 2025</i> Zürich 

Education

	PhD in High Energy Particle Physics Centre for High Energy Physics, Indian Institute of Science Supervisor: Prof. B. Ananthanarayan. Defence date: 9 th December 2022; Degree awarded: 18 th March 2023.	<i>Aug 2019 – Feb 2023</i> Bengaluru 
	MS in Physics Department of Physical Science, Indian Institute of Science Supervisor: Prof. B. Ananthanarayan. Degree awarded (together with PhD): 18 th March 2023.	<i>Aug 2017 – Jul 2019</i> Bengaluru 
	BSc in Physics University of Calcutta Degree awarded: 23 rd June 2017.	<i>Aug 2014 – Jul 2017</i> Kolkata 

Research Fellowships & Grants

	Walter Benjamin Fellowship	2026
	Deutsche Forschungsgemeinschaft €225,334 ($\approx$ \$262,300) • 2 years • <i>Declined</i>	Germany 
	PostDoc.Mobility Fellowship	2025
	Swiss National Science Foundation CHF 136,000 ($\approx$ \$173,100) • 2 years	Switzerland 
	Humboldt Postdoctoral Research Fellowship	2025
	Alexander von Humboldt Foundation €72,000 ($\approx$ \$83,800) • 2 years • <i>Declined</i>	Germany 
	UZH Postdoc Grant	2024
	Universität Zürich CHF 58,560 ($\approx$ \$74,500) • 6 months	Switzerland 
	Senior Research Fellowship	2021
	Ministry of Human Resource Development ₹1,260,000 ($\approx$ \$13,100) • 3 years	India 
	Junior Research Fellowship	2019
	Ministry of Human Resource Development ₹744,000 ($\approx$ \$7,700) • 2 years	India 
	Master's Fellowship	2017
	Ministry of Human Resource Development ₹384,000 ($\approx$ \$4,000) • 2 years	India 

Teaching

	Proseminar Theoretische Physik (Teaching Assistant)	2025
	Universität Zürich Assisted students in preparing seminar talks on various physics topics; graded weekly presentations and understanding of the subject.	Zürich 
	Flavour Physics (Teaching Assistant)	2024
	Universität Zürich Conducted weekly one-hour tutorial classes; assisted in conducting the final oral exam.	Zürich 
	Quantum Field Theory I (Teaching Assistant)	2023
	ETH Zürich Conducted weekly two-hour tutorial classes; graded the final written exam.	Zürich 
	Introductory Physics II (Teaching Assistant)	2020
	Indian Institute of Science Assisted students during weekly three-hour experiments; conducted tutorial classes and the final oral exam; graded the final written exam.	Bangalore 

Research Interests

-  Collider Phenomenology • Beyond the Standard Model • Higgs Physics
-  Multi-loop Feynman Integrals • Hypergeometric Functions • Mellin–Barnes Representation
-  Symbolic & Numerical Computer Algebra • Particle-Physics Software

Publications

Preprints

1. Multiple Mellin-Barnes integrals with polygamma functions [Submitted]
S. Banik, S. Friot [2512.19803]

Journal Articles

2. Multiple Mellin-Barnes integrals with polygamma functions [PRD, 2026]
S. Banik, S. Friot [2512.19803]
3. HyperPrecision: A Mathematica package for High-Precision Numerical Evaluation of Multivariate Hypergeometric Functions [CPC, 2026]
S. Banik, S. Bera [2605.30216]
4. Sunset diagrams with up to three mass scales in chiral perturbation theory: a comparative study of the Mellin-Barnes representation technique [EPJST, 2026]
B. Ananthanarayan, S. Banik, V. Bernard, S. Friot, S. Ghosh, U.-G. Meißner [2512.07727]
5. Two-loop anomalous dimensions for baryon-number-violating operators in SMEFT [JHEP, 2026]
S. Banik, A. Crivellin, L. Naterop, P. Stoffer [2510.08682]
6. Some topics concerning the standard model, Feynman integrals and renormalization group methods: a review of some recent investigations and results [EPJST, 2025]
B. Ananthanarayan, S. Banik, S. Bera, A. B. Das, S. Datta, S. Friot, S. Ghosh, M. S. A. Alam Khan, T. Pathak, R. Sarkar, D. Wyler
7. Correlating EDMs and $A \rightarrow \gamma\gamma$ in the 2HDM in light of the diphoton excesses at 95 GeV and 152 GeV [PRD, 2025]
S. Banik, G. Coloretti, A. Crivellin, H. E. Haber [2412.00523]
8. Growing Evidence for a Higgs Triplet [ChPC, 2025]
A. Crivellin, S. Ashanujjaman, S. Banik, G. Coloretti, S. P. Maharathy, B. Mellado [2404.14492]
9. Anatomy of the Real Higgs Triplet Model [JHEP, 2025]
S. Ashanujjaman, S. Banik, G. Coloretti, A. Crivellin, S. P. Maharathy, B. Mellado [2411.18618]
10. Explaining the $\gamma\gamma + X$ Excesses at ≈ 151.5 GeV via the Drell-Yan Production of a Higgs Triplet [PLB, 2025]
S. Ashanujjaman, S. Banik, G. Coloretti, A. Crivellin, S. P. Maharathy, B. Mellado [2402.00101]

11. Uncovering New Higgses in the LHC Analyses of Differential $t\bar{t}$ Cross Sections [JHEP, 2025]
[2308.07953]
S. Banik, G. Coloretti, A. Crivellin, B. Mellado
12. Explanation of the excesses in associated di-photon production at 152 GeV in 2HDM [JHEP, 2024]
[2407.06267]
S. Banik, A. Crivellin
13. Multiple Mellin-Barnes integrals and triangulations of point configurations [PRD, 2024]
[2309.00409]
S. Banik, S. Friot
14. Renormalization Group Evolution with Scalar Leptoquarks [JHEP, 2023]
[2307.06800]
S. Banik, A. Crivellin
15. $SU(2)_L$ triplet scalar as the origin of the 95 GeV excess? [PRD, 2023]
[2306.15722]
S. Ashanujjaman, S. Banik, G. Coloretti, A. Crivellin, B. Mellado, A. Mulaudzi
16. Asymmetric Di-Higgs Signals of the N2HDM- $U(1)$ [PRD, 2023]
[2303.11351]
S. Banik, A. Crivellin, S. Iguro, T. Kitahara
17. Multiple Mellin-Barnes integrals with straight contours [PRD, 2023]
[2212.11839]
S. Banik, S. Friot
18. FeynGKZ: a Mathematica package for solving Feynman integrals using GKZ Hypergeometric systems [CPC, 2023]
[2211.01285]
B. Ananthanarayan, S. Banik, S. Bera, S. Datta
19. Method of Brackets: Revisiting a technique for calculating Feynman integrals and certain definite integrals [PRD, 2023]
[2112.09679]
B. Ananthanarayan, S. Banik, S. Friot, T. Pathak
20. Multiple Series Representations of N -fold Mellin-Barnes Integrals [PRL, 2021]
[2012.15108]
B. Ananthanarayan, S. Banik, S. Friot, S. Ghosh
21. Massive one-loop conformal Feynman integrals and quadratic transformations of multiple hypergeometric series [PRD, 2021]
[2012.15646]
B. Ananthanarayan, S. Banik, S. Friot, S. Ghosh
22. Double box and hexagon conformal Feynman integrals [PRD, 2020]
[2007.08360]
B. Ananthanarayan, S. Banik, S. Friot, S. Ghosh
23. Quadratic and quartic integrals using the method of brackets [Scientia, 2019]
[1909.00962]
B. Ananthanarayan, S. Banik, S. Datta, T. Pathak

Conference Proceedings

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24. Automated Computation of Multiple Mellin-Barnes Integrals Having Polygamma Functions in Their Integrand [MTTD, 2025]
S. Banik, S. Friot
 25. Indications for New Higgs Bosons [CORFU, 2025]
[2605.04233]
A. Crivellin, S. Ashanujjaman, S. Banik, S. P. Maharathy, G. Coloretti
 26. Analytic Evaluation of Multiple Mellin-Barnes Integrals [LL, 2024]
[2407.20120]
S. Banik, S. Friot

- 27. Automated Evaluation of Feynman Integrals Using GKZ Hypergeometric Systems** [DAE-BRNS, 2024]
B. Ananthanarayan, **S. Banik**, S. Bera, S. Datta
- 28. Geometrical methods for the analytic evaluation of multiple Mellin-Barnes integrals** [MTTD, 2023]
S. Banik, S. Friot [2402.04174]
- 29. Differential $e\mu b\bar{b}$ Cross-Sections and New Higgses at the Electroweak Scale** [TOP, 2023]
S. Banik, G. Coloretti, A. Crivellin, B. Mellado [2312.01458]
- 30. New Higgses at the Electroweak Scale and Differential $t\bar{t}$ Distributions** [DIS, 2024]
S. Banik, G. Coloretti, A. Crivellin, S. Bhattacharya, B. Mellado

Invited Seminars & Conference Talks

- 1. FCC precision calculation WG meeting, CERN, 2026** Geneva 
“Algorithms for Analytic and Numerical evaluation of Multiple Mellin-Barnes integrals”
Invited by: Janusz Gluza and Marco Zaro
- 2. EPP Theory Seminar, SLAC & Stanford University, 2026** Menlo Park 
“Geometrical Techniques to Compute Feynman Integrals”
Invited by: SLAC Theory Group, Stanford University.
- 3. New Ways to Higher Points, HU Berlin, 2025** Berlin 
“Analytic Evaluation of Feynman Integrals using Mellin-Barnes Representation”
Invited by: Burkhard Eden.
- 4. Scalars, University of Warsaw, 2025** Warsaw 
“Indications for New Higgs Boson in Associated Di-Photon Searches”
- 5. Amplitudes Seminar Series, 2025** Bonn 
“Algorithms for Analytic Evaluation of Feynman integrals using MB representation”
Invited by: Florian Loebbert and Claude Duhr.
- 6. BCVSPIN Conference, Tribhuvan and Kathmandu University, 2024** Kathmandu 
“Accumulating Evidence for New Higgs Bosons at the LHC”
- 7. Particle Physics Seminar, KIT, 2024** Karlsruhe 
“Accumulating Evidence for New Higgs Boson from Associated Di-Photon Searches”
Hosted by: Ulrich Nierste.
- 8. HIGGS Conference, University of Uppsala, 2024** Uppsala 
“Indications for New Higgs Boson in Associated Di-Photon Searches”
- 9. HEP Seminar, University of Virginia, 2024** Virginia 
“Accumulating Evidence for New Higgs Boson from Associated Di-Photon Searches”
Hosted by: Julian Heeck.

10. **HEP Seminar, University of Cincinnati, 2024** Cincinnati 
“Accumulating Evidence for New Higgs Boson from Associated Di-Photon Searches”
Hosted by: Joachim Brod.
11. **Pitt PACC Seminar, University of Pittsburgh, 2024** Pittsburgh 
“Accumulating Evidence for New Higgs Boson from Associated Di-Photon Searches”
Hosted by: Tao Han, Amit Bhoonah.
12. **3rd ECFA Workshop on e^+e^- Higgs, Top & Electroweak Factories, 2024** Paris 
“Hints for New Higgs Bosons in Associated Di-Photon Production”
Organized by: G. Marchiori, G. Bernardi, N. Morange.
13. **SPS Annual Meeting, 2024** Zürich 
“Hints for New Higgs Bosons at the LHC”
Organized by: CHIPP, SGN, SSPh, LS².
14. **Workshop on Multi-Higgs Models, 2024** Lisbon 
“Accumulating Evidence for New Higgs Bosons at the LHC”
Organized by: J. Romão, J. P. Silva, R. Santos, P. M. Ferreira, L. Lavoura.
15. **Theoretical Particle Physics Journal Club, 2024** Zürich 
“Recent progress in Evaluating Scalar Feynman Integrals”
Invited by: O. Lara Crosas, C. Savoini and M. Pesut.
16. **LTP(izza)hD, Paul Scherrer Institute, 2024** Villigen 
“Geometrical Approaches to Evaluate Feynman Integrals”
Invited by: S. Kollatzsch and T. D. Hume.
17. **Loops and Legs in Quantum Field Theory, 2024** Wittenberg 
“Analytic Evaluation of Multiple Mellin-Barnes Integrals”
Hosted by: M. Steinhauser and P. Marquard.
18. **High-Energy Physics Group Lunch Seminar, 2023** Chicago 
“Hints for new Higgses at the electroweak scale”
Hosted by: Carlos Wagner and David Miller.
19. **16th Int. Workshop on Top Quark Physics, 2023** Traverse City 
“Uncovering New Higgses in the LHC Analyses of Differential $t\bar{t}$ Cross Sections”
Hosted by: North American Institutes.
20. **Scalars, University of Warsaw, 2023** Warsaw 
“Asymmetric Di-Higgs Signals”
21. **Scalars, University of Warsaw, 2023** Warsaw 
“Triplet Explanation of the 95 GeV Excess”

- 22. CHIPP/CHART Workshop on Sustainability in Particle Physics, 2023** Sursee 
“Asymmetric Di-Higgs Signals of the N^2 HDM”
 Hosted by: Swiss Institute of Particle Physics.
- 23. SPS HEP Seminar, 2023** Bhubaneswar 
“Evaluating Feynman integrals using N -fold Mellin-Barnes integrals”
 Invited by: Narayan Rana.
- 24. CHEP In-House Symposium, Indian Institute of Science, 2022** Bengaluru 
“Triangulations, Mellin-Barnes and Feynman Integrals”
- 25. Elliptic Integrals in Fundamental Physics, 2022** Mainz 
“Hypergeometric solutions of Feynman Integrals using Mellin-Barnes Integrals”
 Invited by: J. Brödel, E. Chaubey, C. Duhr, S. Weinzierl.
- 26. Tropical and Convex Geometry and Feynman Integrals, 2022** Zurich 
“Mellin-Barnes, Conic-Hulls and Triangulations”
 Invited by: Michael Borinsky and Erik Panzer.
- 27. Amplitudes School, 2022** Prague 
“Multi-fold Mellin-Barnes with applications to Feynman integrals”
 Organized by: Karol Kampf and Jaroslav Trnka.
- 28. Joint Bonn-Berlin Seminar (online), 2021** Berlin 
“ N -fold Mellin-Barnes integrals and applications to conformal Feynman integrals and multivariable hypergeometric functions”
 Invited by: Claude Duhr and Florian Loebbert.
- 29. HEP Remote Video Seminar Series (online), 2021** Hyderabad 
“Feynman integrals, Hypergeometric functions and Mellin-Barnes representation”
 Invited by: Anurag Tripathi.
- 30. CHEP In-House Symposium, Indian Institute of Science, 2021** Bengaluru 
“ N -fold Mellin-Barnes Representation and Feynman Integrals”
- 31. CHEP In-House Symposium, Indian Institute of Science, 2020** Bengaluru 
“Decoding the Divergences of the Method of Brackets”

Journal Referee

-  Next Research • Elsevier
-  Physics Letters B • Elsevier
-  Modern Physics Letters A • World Scientific
-  Journal of High Energy Physics • Springer
-  Symmetry, Integrability and Geometry: Methods and Applications • Open Access
-  Mathematics • MDPI

Supervising & Mentoring



Corentin Eggenspieler

ENS Paris-Saclay 

Master's student, 2025

Thesis: “*Novel Computation Techniques for Perturbative Quantum Field Theory*”.



Banurega Shanmuganathan

Universität Zürich 

Bachelor's student, 2024

Thesis: “*Compendium of Two-Body Decays*”.



Mathieu Ferey

Université Paris-Saclay 

Master's student, 2022

Thesis: “*Computation of Feynman Diagrams and their transformation into Mellin-Barnes integrals for Quantum Field Theory*”.

Programming Skills & Interests

Languages

Wolfram Language

Python

Fortran

Java

C++

ROOT

Software & HEP Tools

MadGraph

Pythia

Delphes

MadAnalysis

SARAH

SPheno

FeynRules

FeynArts

FeynCalc

FIESTA

AMBRE

MB

TopCom

Kira

FIRE

Software Developed

 MBConicHulls

 FeynGKZ

 HyperPrecision

Personal Interests

 Computer Programming

 History

 Cricket

 Chess

 Travelling